

HbA1c

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What is HbA1c? | Blood Test | Target Levels | Diabetes UK

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What is HbA1c?

HbA1c is your average blood glucose (sugar) levels for the last two to three months. If you have diabetes, an ideal HbA1c level is 48mmol/mol (6.5%) or below.

If you're at risk of developing type 2 diabetes , your target HbA1c level should be below 42mmol/mol (6%).

On this page we'll go through what HbA1c means, and why aiming for your target level is so important. We'll also explain the HbA1c test in more detail, and show you what you can do to lower your HbA1c levels if they're too high. Plus, you can use our HbA1c converter tool if you're looking to find out your level in % or mmol/mol.

What does HbA1c mean?

HbA1c is what's known as glycated haemoglobin. This is something that's made when the glucose (sugar) in your body sticks to your red blood cells. Your body can't use the sugar properly, so more of it sticks to your blood cells and builds up in your blood. Red blood cells are active for around 2-3 months, which is why the reading is taken quarterly.

A high HbA1c means you have too much sugar in your blood. This means you're more likely to develop diabetes complications , like serious problems with your eyes and feet.

Knowing your HbA1c level and what you can do to lower it will help you reduce your risk of devastating complications. This means getting your HbA1c checked regularly. It's a vital check and part of your annual review . You're entitled to get this test at least once a year. But if your HbA1c is high or needs a little more attention, it'll be done every three to six months.

It's really important not to skip your diabetes health checks, so if you haven't had an HbA1c test in over a year contact your healthcare team.

Once you know your HbA1c level, it's important that you understand what the results mean and how to stop them from getting too high. Even a slightly raised HbA1c level makes you more at risk of serious complications, so get all the facts here and be in the know about HbA1c.

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What is a HbA1c test?

The haemoglobin A1c (HbA1c) test measures the amount of blood sugar (glucose) attached to your haemoglobin. Haemoglobin is the part of your red blood cells that carries oxygen from your lungs to the rest of your body. It is an important blood test that gives a good indication of how well your diabetes is being managed.

You can check these average blood sugar levels yourself, but you'll have to buy a kit, whereas your healthcare professional will do it for free. It's different from a finger-prick test, which is a snapshot of your blood sugar levels at a particular time, on a particular day.

You find out your HbA1c level by getting a blood test by a doctor or nurse. Your healthcare team will arrange this for you, but chase it up with your GP if you haven't had one for a few months.

The accuracy of the HbA1c test can be affected by other medical conditions especially conditions that affect the red blood cells such as haemolytic anaemia, sickle cell anaemia and thalassaemia. If you have a condition that affects your red blood cells talk to your doctor about what other test will be more accurate for you. Most people will have the test every three to six months. But you may need it more often if you're planning for a baby, your treatment has recently changed, or you're having problems managing your blood sugar levels.

And some people will need the test less often, usually later on during pregnancy.

An HbA1c test is also used to diagnose diabetes, and to keep an eye on your levels if you're at risk of developing diabetes (you have prediabetes).

The test is sometimes called haemoglobin A1c or just A1c.

Your HbA1c results

You should get the results quickly. The result of the HbA1c test lets your healthcare team know if they need to change your treatment or medication to help you manage your levels better. But it also tells you a number and it's important you understand what this means. Some people find it helps to write their results down in a diary, to keep track of them and see if they can spot any trends.

And your HbA1c can change for lots of reasons, including:

if you're unwell if you're taking other medicines, like steroids changes in lifestyle if you're feeling very stressed or you're depressed.

You'll talk to your healthcare team about a target level that you need to aim for. We know this isn't as easy as it sounds, but it's important you do everything you can to keep in your target range. The longer your HbA1c level is even slightly high, the more you're at risk of developing life-threatening complications.

We have lots of information and tools to help you lower your HbA1c.

What is a normal range for an HbA1c test?

If you have diabetes, an ideal HbA1c level is 48mmol/mol (6.5%) or below.

HbA1c levels and targets

But everyone's different. So your healthcare team may give you an individual target level that takes into account your current level and when your next test is. So you can bring it down in stages and isn't a sudden big drop.

And we're not saying it's easy to get to and stay at this level. We know it's difficult, and we're here to give you advice and information to help.

If you're at risk of type 2 diabetes

There are different target HbA1c levels for people at risk of developing type 2 diabetes. If you've been told you're at risk, your target level should be below 42mmol/mol (6%). We have lots more information for you if

you're at risk of type 2 .

Type 2 diabetes remission

Remission is when a person with type 2 diabetes has healthy blood glucose (also called sugar) levels for the long-term, without taking any diabetes medications. We're working with international experts to agree this, but our researchers used an HbA1c level of 48mmol/mol (6.5%) or less to define remission.

Type 2 diabetes is still a serious condition. It can be lifelong and get worse over time for many, but it doesn't have to be like this for everyone. This can be life-changing. Find out more about type 2 diabetes remission .

Older people

As people age, their target HbA1c may be increased by their diabetes healthcare team. This is because the risks of keeping your HbA1c to a target of 48mmol/mol may outweigh the benefits. For example, there may be greater risk of a harm from a fall due to a low blood sugar than the risk of long term health problems linked to higher blood sugar levels. If you are unsure if your target have changed or why your targets have changed you should ask your diabetes team about it at your next appointment.

HbA1c converter

If you're wondering how to convert HbA1c mmol/mol to %, or vice versa, our tool can help you.

Mmol/mol stands for millimoles per mole. A mole is a scientific unit often used to measure chemicals, and it has been the standard measurement for glucose levels since 2009. Before that, a percentage was used. That's why measurements often have a percentage as well.

If you're more used to the percentage system, use our converter tool below to find out your level in millimoles.

Convert HbA1c % to mmol/mol and vice versa

Select the value for your conversion: convert % to mmol/mol convert mmol/mol to % Enter value:

The HbA1c calculator was developed by ©Diabetes UK Wells Lawrence House, 126 Back Church Lane, London E1 1FH. www.diabetes.org.uk

How to lower your HbA1c levels

If your levels have gone above your target since your last check, it's understandable to be worried. Even a slightly high HbA1c level puts you more at risk of developing serious complications in your body. But knowing your numbers and what that means is an important and good first step – now you need to know how to lower them.

Lots of things can cause your HbA1c levels to change and there's action you and your healthcare team can take to bring them down to your target level:

Your diabetes team may need to review your medication and increase the dose or try a new one. Ask them about local diabetes education courses – there's always more to learn with diabetes and a course will help you take practical steps to lower your HbA1c. Get more active – moving more is good for everyone, but it can specifically help to bring down your HbA1c levels. Get advice on balanced, healthy eating . Stop smoking – smoking makes it harder for blood to flow around your body.

And if you want to hear from others who are experiencing this too, think about joining our online forum . It's a place where you can chat to others or just read other people's stories.

And don't forget, get advice from your healthcare team . They're here to help.

HbA1c test and finger-prick test – what's the difference?

Get more information on checking your own blood sugars using a finger-prick test.

HbA1c test Finger-prick test Measures your average blood sugar levels over last 2 to 3 months Snapshot of blood sugar levels at the time Everyone with diabetes should have this test Only people taking certain medicines need to do this test, like insulin and sulphonylureas Usually done every 3 to 6 months Done a number of times a day, usually ranging from twice to 10 times Blood is taken from arm or finger Tiny drop of blood taken from the finger A healthcare professional should do this for you, though you can pay for a kit and do it yourself. These aren't available on prescription. You can do this yourself, using a testing kit Doctors can use this test to find out how someone is managing their diabetes, to diagnose diabetes and find out if someone is at risk of type 2 diabetes This test is mainly used for people living with diabetes

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